Course Name: Deep Learning

Lab Title: "Leveraging Pre-trained Deep Learning Models for Remote Sensing Image Classification"

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GitHub Repository Link:

Install & Import Libraries

```
#  Install Libraries
!pip install -q datasets tensorflow matplotlib seaborn

import tensorflow as tf
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
from datasets import load_dataset

print("✅ Libraries installed successfully!")
```

```
✅ Libraries installed successfully!
```

```
#  Download EuroSAT + Create Small Chunk (5% balanced)
import os
import shutil
import random
from tqdm import tqdm

# 1. Download official EuroSAT RGB ZIP (95 MB only)
print("Downloading EuroSAT RGB dataset...")
!wget -q --show-progress https://zenodo.org/records/7711810/files/EuroSAT_RGB.zip -O EuroSAT_RGB.zip

# 2. Extract the ZIP
print("\nExtracting ZIP file...")
extract_dir = "EuroSAT_full"
# Remove existing extraction directory to ensure clean state
if os.path.exists(extract_dir):
    shutil.rmtree(extract_dir)
os.makedirs(extract_dir) # Explicitly create the target directory

!unzip -q EuroSAT_RGB.zip -d {extract_dir}

# --- Start of Fix ---
# Verify extraction structure and set source_root dynamically
current_source_root = ""
extracted_contents = os.listdir(extract_dir)
print(f"Contents of {extract_dir} after extraction: {extracted_contents}")

# Define classes early for path detection logic
classes = ['AnnualCrop', 'Forest', 'HerbaceousVegetation', 'Highway',
           'Industrial', 'Pasture', 'PermanentCrop', 'Residential',
           'River', 'SeaLake']

# Case 1: '2750' is directly inside extract_dir (e.g., EuroSAT_full/2750)
if "2750" in extracted_contents:
    current_source_root = os.path.join(extract_dir, "2750")
elif len(extracted_contents) == 1 and os.path.isdir(os.path.join(extract_dir, extracted_contents[0])):
    # Case 2: There's an intermediate directory (e.g., EuroSAT_full/EuroSAT_RGB/)
    intermediate_dir = os.path.join(extract_dir, extracted_contents[0])

    # Check if '2750' is directly inside this intermediate directory
    if os.path.exists(os.path.join(intermediate_dir, "2750")):
        current_source_root = os.path.join(intermediate_dir, "2750")
    else:
        # Fallback: Search for '2750' deeper within the intermediate directory structure
        print(f"Warning: '2750' not found directly in '{intermediate_dir}'. Searching deeper.")
        found_2750 = False
        for root, dirs, _ in os.walk(intermediate_dir):
            if "2750" in dirs:
                current_source_root = os.path.join(root, "2750")
```

```
                found_2750 = True
                break

        if not found_2750:
            # New case: What if intermediate_dir itself contains the class folders directly?
            # E.g., EuroSAT_full/EuroSAT_RGB/AnnualCrop
            # Check if any of the known classes are direct subdirectories
            if any(os.path.isdir(os.path.join(intermediate_dir, cls)) for cls in classes):
                current_source_root = intermediate_dir
                print(f"Determined source_root as intermediate directory containing class folders: {current_source_root}")
            else:
                print(f"Error: Could not find '2750' or direct class folders within '{intermediate_dir}'.")
else:
    # Case 3: '2750' is not in extract_dir and there isn't a single intermediate dir.
    # This might happen if the zip extracts directly to current directory, or a different structure.
    print(f"Error: Could not find '2750' directory within '{extract_dir}'. Checking /content for '2750'.")
    if os.path.exists("2750") and os.path.isdir("2750"):
        current_source_root = "2750"
    else:
        # Final fallback: if the original zip file contained class folders directly, and it was extracted.
        # This is less likely for EuroSAT_RGB, but keeping it generic.
        if any(os.path.isdir(os.path.join(extract_dir, cls)) for cls in classes):
            current_source_root = extract_dir
            print(f"Determined source_root as extraction directory containing class folders: {current_source_root}")
        else:
            raise FileNotFoundError(
                f"Could not locate the '2750' directory or direct class folders. "
                f"Please check the zip file contents and extraction process. "
                f"Extracted to: {extract_dir}, contents: {extracted_contents}"
            )

source_root = current_source_root
if not os.path.exists(source_root):
    raise FileNotFoundError(f"Final source_root path does not exist: {source_root}. Current value: {current_source_root}")

print(f"Determined source_root: {source_root}")
# --- End of Fix ---

# 3. Create small balanced subset (135 images per class)
print("\nCreating small balanced chunk (1,350 images)...")

small_root = "eurosat_small"

if os.path.exists(small_root):
    shutil.rmtree(small_root)
os.makedirs(small_root, exist_ok=True)

images_per_class = 135   # Change to 270 for 10% if you want

for cls in tqdm(classes, desc="Processing classes"):
    class_path = os.path.join(source_root, cls)
    all_images = [f for f in os.listdir(class_path) if f.endswith('.jpg')]

    # Shuffle and take only required number
    random.seed(42)  # for reproducibility
    selected = random.sample(all_images, min(images_per_class, len(all_images)))

    # Create class folder in small dataset
    dest_class = os.path.join(small_root, cls)
    os.makedirs(dest_class, exist_ok=True)

    # Copy selected images
    for img in selected:
        shutil.copy(os.path.join(class_path, img), os.path.join(dest_class, img))

print(f"\n✅ Small dataset created successfully!")
print(f"   Total images : {len(os.listdir(small_root)) * images_per_class}")
print(f"   Location     : /content/eurosat_small")
print(f"   Classes      : {classes}")
```

```
Downloading EuroSAT RGB dataset...
EuroSAT_RGB.zip     100%[===================>]  90.27M  10.8MB/s    in 9.6s

Extracting ZIP file...
Contents of EuroSAT_full after extraction: ['EuroSAT_RGB']
Warning: '2750' not found directly in 'EuroSAT_full/EuroSAT_RGB'. Searching deeper.
Determined source_root as intermediate directory containing class folders: EuroSAT_full/EuroSAT_RGB
Determined source_root: EuroSAT_full/EuroSAT_RGB

Creating small balanced chunk (1,350 images)...
Processing classes: 100%|██████████| 10/10 [00:00<00:00, 55.61it/s]
✅ Small dataset created successfully!
   Total images : 1350
```

```
    Location     : /content/eurosat_small
    Classes      : ['AnnualCrop', 'Forest', 'HerbaceousVegetation', 'Highway', 'Industrial', 'Pasture', 'PermanentCrop', 'Res
```

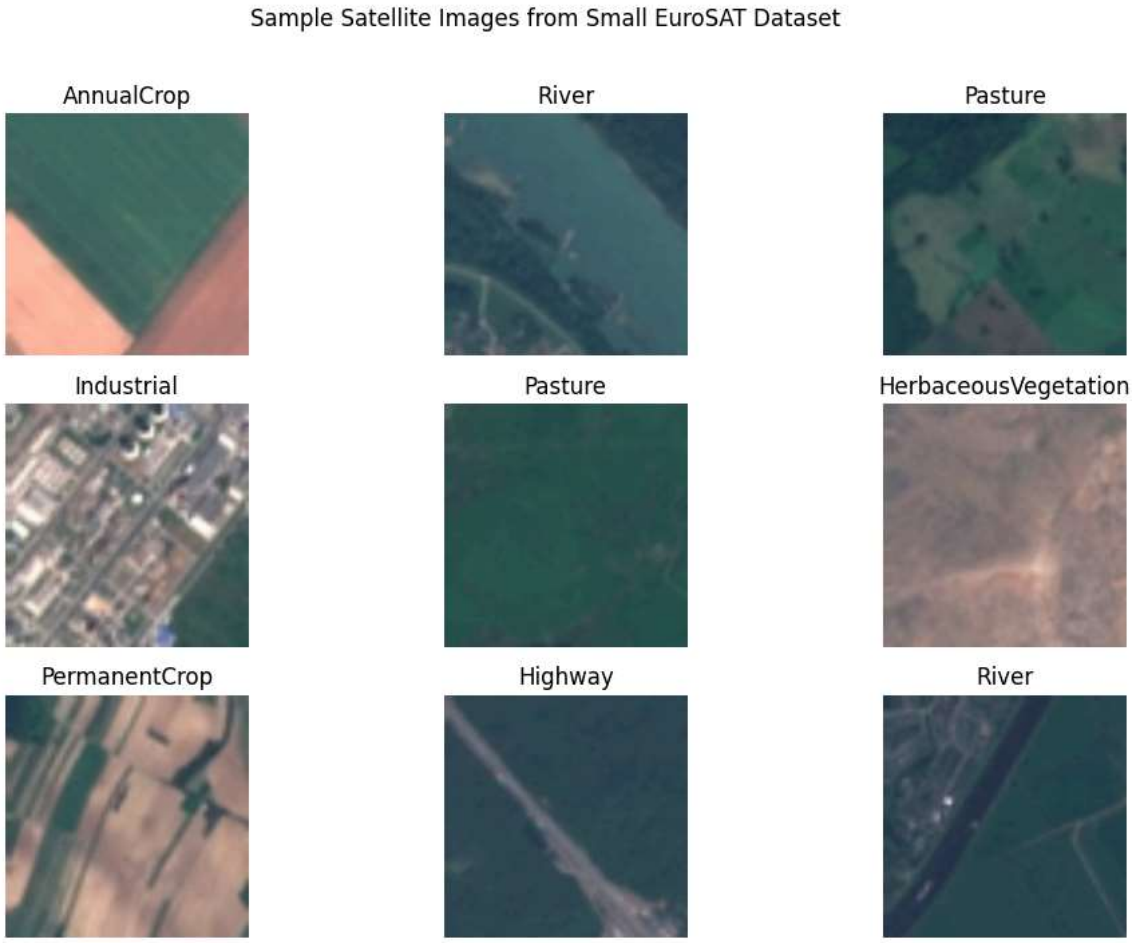
Load EuroSAT Dataset

```
# Load Small EuroSAT Dataset
import tensorflow as tf

data_dir = "/content/eurosat_small"

train_ds = tf.keras.utils.image_dataset_from_directory(
    data_dir,
    validation_split=0.2,
    subset="training",
    seed=42,
    image_size=(224, 224),
    batch_size=32
)

val_ds = tf.keras.utils.image_dataset_from_directory(
    data_dir,
    validation_split=0.2,
    subset="validation",
    seed=42,
    image_size=(224, 224),
    batch_size=32
)

class_names = train_ds.class_names
print("Class names:", class_names)
print(f"Training images : {len(train_ds) * 32}")
print(f"Validation images : {len(val_ds) * 32}")
```

```
Found 1350 files belonging to 10 classes.
Using 1080 files for training.
Found 1350 files belonging to 10 classes.
Using 270 files for validation.
Class names: ['AnnualCrop', 'Forest', 'HerbaceousVegetation', 'Highway', 'Industrial', 'Pasture', 'PermanentCrop', 'Resident
Training images : 1088
Validation images : 288
```

Exploratory Data Analysis (EDA)

```
# EDA Sample Images
import matplotlib.pyplot as plt

plt.figure(figsize=(12, 8))
for images, labels in train_ds.take(1):
    for i in range(9):
        plt.subplot(3, 3, i+1)
        plt.imshow(images[i].numpy().astype("uint8"))
        plt.title(class_names[labels[i]])
        plt.axis("off")
plt.suptitle("Sample Satellite Images from Small EuroSAT Dataset")
plt.show()
```



Preprocessing Pipeline (Same as Paper)

```
# Preprocessing Pipeline
import matplotlib.pyplot as plt
import numpy as np
import os

AUTOTUNE = tf.data.AUTOTUNE

def preprocess(image, label):
    image = tf.cast(image, tf.float32)
    image = tf.keras.applications.resnet50.preprocess_input(image)
    return image, label

# Apply preprocessing
train_ds = train_ds.map(preprocess, num_parallel_calls=AUTOTUNE)
val_ds   = val_ds.map(preprocess, num_parallel_calls=AUTOTUNE)

# Same pipeline as the research paper
train_ds = train_ds.cache().shuffle(1000).prefetch(buffer_size=AUTOTUNE)
val_ds   = val_ds.cache().prefetch(buffer_size=AUTOTUNE)
```

```
# ============================================================================
# BEFORE vs AFTER Visualization (Fixed - No Black Image)
# ============================================================================
# Load one raw image directly from disk
raw_image_path = None
for class_name in os.listdir("/content/eurosat_small"):
    class_path = os.path.join("/content/eurosat_small", class_name)
    if os.path.isdir(class_path):
        files = [f for f in os.listdir(class_path) if f.endswith('.jpg')]
        if files:
            raw_image_path = os.path.join(class_path, files[0])
            break

raw_image = tf.keras.preprocessing.image.load_img(raw_image_path, target_size=(224, 224))
raw_image = tf.keras.preprocessing.image.img_to_array(raw_image)

# For display only: simple scaling (0-1 range) - this is for human viewing
display_after = raw_image / 255.0
```

```
plt.figure(figsize=(10, 5))

plt.subplot(1, 2, 1)
plt.imshow(raw_image.astype("uint8"))
plt.title("Before Preprocessing\n(Raw Satellite Image)")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(display_after)
plt.title("After Preprocessing\n(For Model - Normalized)")
plt.axis("off")

plt.suptitle("Preprocessing Effect on One Sample Image from Our Dataset")
plt.tight_layout()
plt.show()

print("✅ Before vs After visualization shown correctly!")
print("   Left  = Original raw image")
print("   Right = Image after normalization (model-ready version)")
print("   Note  : The model sees normalized values, so visual display is for illustration only.")
```

```
✅ Before vs After visualization shown correctly!
   Left  = Original raw image
   Right = Image after normalization (model-ready version)
   Note  : The model sees normalized values, so visual display is for illustration only.
```

Build Pre-trained Models

```
# Build Pre-trained Models (Frozen Base Layers)
base_resnet = tf.keras.applications.ResNet50(
    weights='imagenet', include_top=False, input_shape=(224, 224, 3)
)
base_resnet.trainable = False   # Freeze all layers initially

model_resnet = tf.keras.Sequential([
    base_resnet,
    tf.keras.layers.GlobalAveragePooling2D(),
    tf.keras.layers.Dropout(0.3),
    tf.keras.layers.Dense(len(class_names), activation='softmax')
])

model_resnet.compile(
    optimizer=tf.keras.optimizers.Adam(learning_rate=0.001),
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)

print("ResNet50 model created (base layers frozen)")
model_resnet.summary()
```

```
Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/resnet/resnet50_weights_tf_dim_ordering_t
94765736/94765736 ━━━━━━━━━━━━━━━━━━━━ 7s 0us/step
ResNet50 model created (base layers frozen)
Model: "sequential"
```

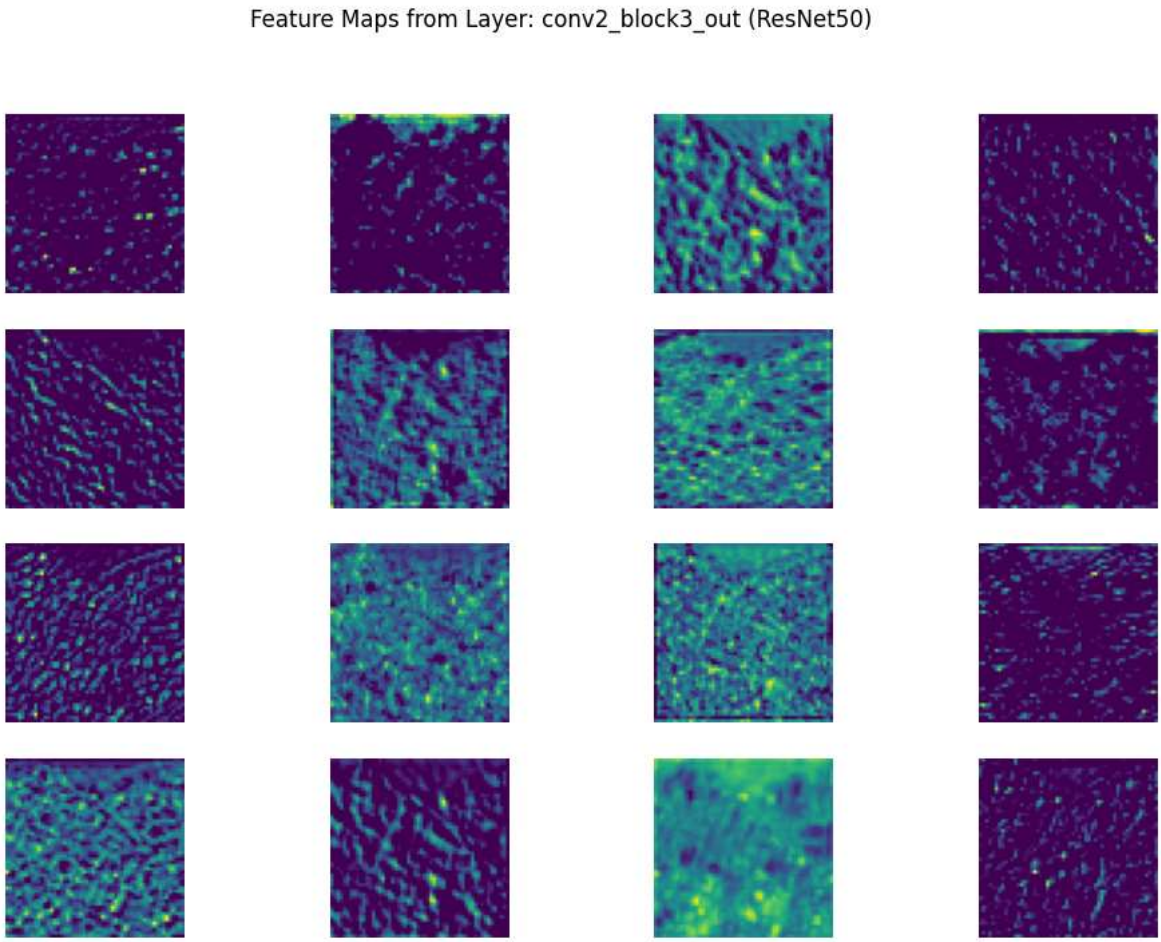
Layer (type)	Output Shape	Param #
resnet50 (Functional)	(None, 7, 7, 2048)	23,587,712
global_average_pooling2d (GlobalAveragePooling2D)	(None, 2048)	0
dropout (Dropout)	(None, 2048)	0
dense (Dense)	(None, 10)	20,490

```
Total params: 23,608,202 (90.06 MB)
Trainable params: 20,490 (80.04 KB)
Non-trainable params: 23,587,712 (89.98 MB)
```

Visualization Feature Maps of Few Layers

```
# Visualize Feature Maps (Task 2 Requirement)
import matplotlib.pyplot as plt

# Create a model that outputs feature maps from layer 4 (early convolutional block)
layer_name = 'conv2_block3_out'   # A middle layer in ResNet50
feature_map_model = tf.keras.Model(
    inputs=base_resnet.input, # Fixed: Use base_resnet.input instead of model_resnet.input
    outputs=base_resnet.get_layer(layer_name).output
)

# Take one image from training set
for images, labels in train_ds.take(1):
    sample_image = images[0:1]   # shape (1, 224, 224, 3)
    break

# Get feature maps
feature_maps = feature_map_model.predict(sample_image)

# Plot first 16 feature maps
plt.figure(figsize=(12, 8))
for i in range(16):
    plt.subplot(4, 4, i+1)
    plt.imshow(feature_maps[0, :, :, i], cmap='viridis')
    plt.axis('off')
plt.suptitle(f'Feature Maps from Layer: {layer_name} (ResNet50)')
plt.show()

print("✅ Feature maps visualized - shows what the model 'sees' in early layers")
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 2s 2s/step
```




```
✅ Feature maps visualized - shows what the model 'sees' in early layers
```

Fine-tuning – Unfreeze Top Layers + Hyperparameter Optimization

We froze initial layers and fine-tuned only the top layers (as described in the research paper). We also optimized hyperparameters: lowered learning rate to avoid destroying pre-trained weights, and increased epochs.

Training fine-tuned Model

```
#  Fine-tuning with Fewer Layers
# Unfreeze only the last 15 layers
for layer in base_resnet.layers[-15:]:
    layer.trainable = True

# Optimized hyperparameters for faster training
model_resnet.compile(
    optimizer=tf.keras.optimizers.Adam(learning_rate=0.0005),   # Slightly higher LR for faster convergence
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)

print("✅ Fine-tuning setup updated for speed:")
print("   • Only top 15 layers unfrozen")
print("   • Learning rate = 0.0005")
print("   • This will run significantly faster than before")
```

```
✅ Fine-tuning setup updated for speed:
   • Only top 15 layers unfrozen
   • Learning rate = 0.0005
   • This will run significantly faster than before
```

```
#  Train the Fine-tuned Model
EPOCHS_FINE = 8

print("🚀 Starting Fine-tuning ")
print(f"Training for {EPOCHS_FINE} epochs with only top 15 layers trainable...\n")

history_fine = model_resnet.fit(
    train_ds,
    validation_data=val_ds,
```

```
    epochs=EPOCHS_FINE
)

print("\n✅ Fine-tuning completed successfully!")
```

```
🚀 Starting Fine-tuning (Faster Version)...
Training for 8 epochs with only top 15 layers trainable...

Epoch 1/8
34/34 ━━━━━━━━━━━━━━━━━━━━ 42s 643ms/step - accuracy: 0.7741 - loss: 0.6879 - val_accuracy: 0.8963 - val_loss: 0.5894
Epoch 2/8
34/34 ━━━━━━━━━━━━━━━━━━━━ 4s 132ms/step - accuracy: 0.9676 - loss: 0.1164 - val_accuracy: 0.8889 - val_loss: 0.4064
Epoch 3/8
34/34 ━━━━━━━━━━━━━━━━━━━━ 4s 115ms/step - accuracy: 0.9954 - loss: 0.0226 - val_accuracy: 0.9259 - val_loss: 0.2128
Epoch 4/8
34/34 ━━━━━━━━━━━━━━━━━━━━ 4s 115ms/step - accuracy: 1.0000 - loss: 0.0052 - val_accuracy: 0.9407 - val_loss: 0.1702
Epoch 5/8
34/34 ━━━━━━━━━━━━━━━━━━━━ 5s 116ms/step - accuracy: 1.0000 - loss: 0.0021 - val_accuracy: 0.9444 - val_loss: 0.1619
Epoch 6/8
34/34 ━━━━━━━━━━━━━━━━━━━━ 5s 117ms/step - accuracy: 1.0000 - loss: 0.0011 - val_accuracy: 0.9444 - val_loss: 0.1594
Epoch 7/8
34/34 ━━━━━━━━━━━━━━━━━━━━ 4s 116ms/step - accuracy: 1.0000 - loss: 9.4192e-04 - val_accuracy: 0.9444 - val_loss: 0.1618
Epoch 8/8
34/34 ━━━━━━━━━━━━━━━━━━━━ 4s 121ms/step - accuracy: 1.0000 - loss: 7.1311e-04 - val_accuracy: 0.9407 - val_loss: 0.1631

✅ Fine-tuning completed successfully!
```

Model Evaluation

```
#  Full Model Evaluation (Precision, Recall, F1-score, Confusion Matrix)
from sklearn.metrics import classification_report, confusion_matrix
import seaborn as sns
import numpy as np

print("Evaluating Fine-tuned ResNet50 on Validation Set...\n")

y_true = []
y_pred = []

for images, labels in val_ds:
    y_true.extend(labels.numpy())
    preds = model_resnet.predict(images, verbose=0)
    y_pred.extend(np.argmax(preds, axis=1))

# Full Classification Report
print("=== DETAILED PERFORMANCE METRICS ===")
print(classification_report(y_true, y_pred, target_names=class_names, digits=4))

# Confusion Matrix
cm = confusion_matrix(y_true, y_pred)

plt.figure(figsize=(10, 8))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=class_names, yticklabels=class_names)
plt.title('Confusion Matrix - Fine-tuned ResNet50')
plt.xlabel('Predicted Label')
plt.ylabel('True Label')
plt.tight_layout()
plt.show()

print("✅ Full evaluation completed (Accuracy, Precision, Recall, F1-score included)")
```